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SECTION:	S-22
SUBJECT:	FULL STACK APPLICATION DEVELOPMENT



#SKILL - 1 Git Version Control

Tasks:

1. Create a new project folder, initialize Git using git init, and configure Git with your username and email.

Commands Used:-

```
mkdir MyGitProject
cd MyGitProject
```

```
git init
git config user.name "Your Name"
git config user.email "youremail@example.com"
```

Output:-

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\lalith>E:

E:\>cd "Full Stack Development"

E:\Full Stack Development>cd "workspace - vsc"

E:\Full Stack Development\workspace - vsc>mkdir "workspace - vsc"

E:\Full Stack Development\workspace - vsc>|
```

2. Create two files (example: main.java, notes.txt) with sample content, check the repository status, and add them to staging.

Commands Used:-

```
echo public class Main { public static void main(String[] args) { System.out.println("Hello
Git!"); } } > main.java
echo This is my first Git project. > notes.txt
```

```
git status
git add main.java notes.txt
```

Output:-

```
E:\Full Stack Development\workspace - vsc\Skill Experiments>git branch -M main
E:\Full Stack Development\workspace - vsc\Skill Experiments>git remote add origin https://github.com/lalithdev/2400031810-FSAD20252026EVENSEM-SKILL-EXPERIMENTS.git
E:\Full Stack Development\workspace - vsc\Skill Experiments>git push -u origin main
Enumerating objects: 274, done.
Counting objects: 100% (274/274), done.
Delta compression using up to 12 threads
Compressing objects: 100% (194/194), done.
Writing objects: 100% (274/274), 4.23 MiB | 771.00 KiB/s, done.
Total 274 (delta 35), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (35/35), done.
To https://github.com/lalithdev/2400031810-FSAD20252026EVENSEM-SKILL-EXPERIMENTS.git
 * [new branch]    main -> main
branch 'main' set up to track 'origin/main'.
E:\Full Stack Development\workspace - vsc\Skill Experiments>
```

3. Commit the staged files with a meaningful commit message.

Commands Used:-

git commit -m "Initial commit: add main.java and notes.txt"

Output:-

```
E:\Full Stack Development\workspace - vsc\Skill Experiments>git branch -M main
E:\Full Stack Development\workspace - vsc\Skill Experiments>git remote add origin https://github.com/lalithdev/2400031810-FSAD20252026EVENSEM-SKILL-EXPERIMENTS.git
E:\Full Stack Development\workspace - vsc\Skill Experiments>git push -u origin main
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To https://github.com/lalithdev/2400031810-FSAD20252026EVENSEM-SKILL-EXPERIMENTS.git
 * [new branch]    main -> main
branch 'main' set up to track 'origin/main'.
E:\Full Stack Development\workspace - vsc\Skill Experiments>
```

4. Create a new repository on GitHub, connect your local repo using the remote URL and push the committed changes.

Commands Used:-

git branch -M main

git remote add origin https://github.com/USERNAME/MyGitProject.git

git push -u origin main

Output:-

```
E:\Full Stack Development\workspace - vsc\Skill Experiments>git branch -M main
E:\Full Stack Development\workspace - vsc\Skill Experiments>git remote add origin https://github.com/lalithdev/2400031810-FSAD20252026EVENSEM-SKILL-EXPERIMENTS.git
E:\Full Stack Development\workspace - vsc\Skill Experiments>git push -u origin main
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To https://github.com/lalithdev/2400031810-FSAD20252026EVENSEM-SKILL-EXPERIMENTS.git
 * [new branch]    main -> main
branch 'main' set up to track 'origin/main'.
E:\Full Stack Development\workspace - vsc\Skill Experiments>
```

5. Create the first branch (example: feature-update), make modifications, and add + commit the changes.

Commands Used:-

```
git checkout -b feature-update
echo Added feature update notes. >> notes.txt
git add notes.txt
git commit -m "Add feature update notes"
```

Output:-

```
E:\Full Stack Development\workspace - vsc\Skill Experiments>git branch -M main
E:\Full Stack Development\workspace - vsc\Skill Experiments>git remote add origin https://github.com/lalithdev/2400031810-FSAD20252026EVENSEM-SKILL-EXPERIMENTS.git
E:\Full Stack Development\workspace - vsc\Skill Experiments>git push -u origin main
Enumerating objects: 274, done.
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To https://github.com/lalithdev/2400031810-FSAD20252026EVENSEM-SKILL-EXPERIMENTS.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
E:\Full Stack Development\workspace - vsc\Skill Experiments>
```

6. Create a second branch (example: bug-fix), apply changes, and add + commit them.

Commands Used:-

```
git checkout main
git checkout -b bug-fix
echo System.out.println("Bug fixed!"); >> main.java
git add main.java
git commit -m "Fix bug in main.java"
```

Output:-

```
E:\Full Stack Development\workspace - vsc\Skill Experiments>git branch -M main
E:\Full Stack Development\workspace - vsc\Skill Experiments>git remote add origin https://github.com/lalithdev/2400031810-FSAD20252026EVENSEM-SKILL-EXPERIMENTS.git
E:\Full Stack Development\workspace - vsc\Skill Experiments>git push -u origin main
Enumerating objects: 274, done.
Counting objects: 100% (274/274), done.
Delta compression using up to 12 threads
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Writing objects: 100% (274/274), 4.23 MiB | 771.00 KiB/s, done.
Total 274 (delta 35), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (35/35), done.
To https://github.com/lalithdev/2400031810-FSAD20252026EVENSEM-SKILL-EXPERIMENTS.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
E:\Full Stack Development\workspace - vsc\Skill Experiments>
```

7. Switch back to the main branch and merge both branches one after the other.

Commands Used:-

```
git checkout main
git merge feature-update
git merge bug-fix
```

Output:-

```
E:\Full Stack Development\workspace - vsc\Skill Experiments>git branch -M main
E:\Full Stack Development\workspace - vsc\Skill Experiments>git remote add origin https://github.com/lalithdev/2400031810-FSAD20252026EVENSEM-SKILL-EXPERIMENTS.git
E:\Full Stack Development\workspace - vsc\Skill Experiments>git push -u origin main
Enumerating objects: 274, done.
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To https://github.com/lalithdev/2400031810-FSAD20252026EVENSEM-SKILL-EXPERIMENTS.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
E:\Full Stack Development\workspace - vsc\Skill Experiments>
```

8. If any merge conflict occurs, resolve it manually, commit the fix, and complete the merge

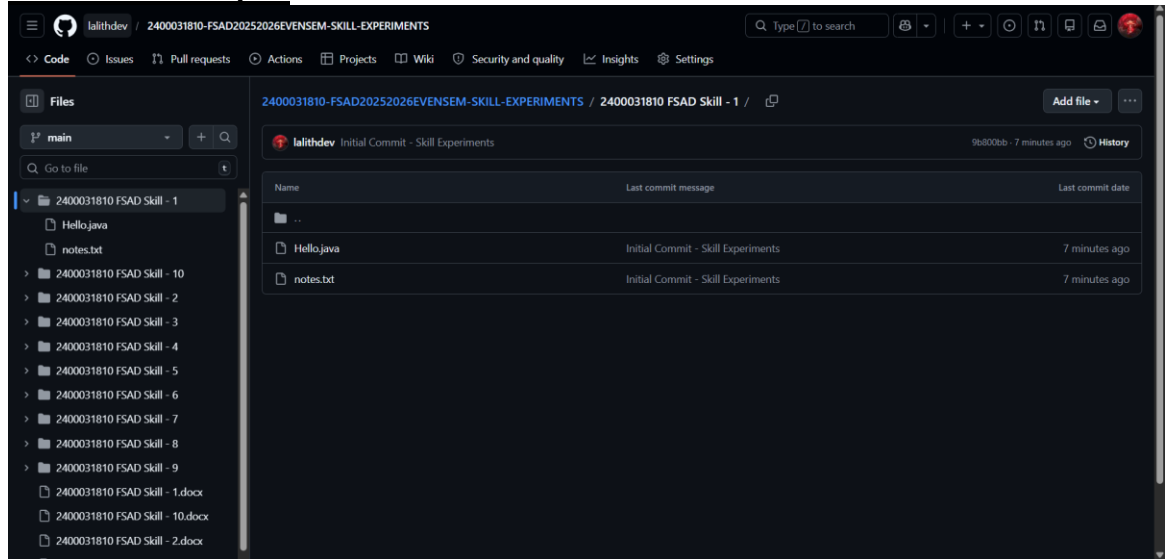
Commands Used:-

```
git add .
git commit -m "Resolve merge conflict"
```

Output:-

```
E:\Full Stack Development\workspace - vsc\Skill Experiments>git branch -M main
E:\Full Stack Development\workspace - vsc\Skill Experiments>git remote add origin https://github.com/lalithdev/2400031810-FSAD20252026EVENSEM-SKILL-EXPERIMENTS.git
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 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
E:\Full Stack Development\workspace - vsc\Skill Experiments>
```

Final GitHub Repo:-



Git Hub link: <https://github.com/lalithdev/2400031810-FSAD20252026EVENSEM-SKILL-EXPERIMENTS.git>

Conclusion:-

In this experiment, Git was successfully used as a version control system to manage a project efficiently. The repository was initialized, user configuration was completed, and files were created, staged, and committed following the Git life cycle. Multiple branches were created to handle feature updates and bug fixes independently, ensuring safe and organized development. These branches were later merged into the main branch, demonstrating effective branch management and merge operations. Finally, the project was pushed to a remote GitHub repository, enabling centralized storage and version tracking. This exercise provided hands-on experience with essential Git commands and workflows used in real-world software development.